DAY 34 | POST-READ

LLM Deployment: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Hosting	Hosted API, cloud-hosted in your tenant, self-hosted open model.
Managed agent runtimes (NEW v4)	Provider runs the loop, tools and state; check lock-in, region and trace export.
Data retention (NEW v4)	Ask the provider, runtime, your logs and memory stores how long data lives.
Versioning and rollback	Prompts, datasets and embeddings in one manifest; rollback = previous manifest.
Environments and secrets	Dev, test, prod; same image; secrets from Key Vault; fail fast if missing.
AI gateway (NEW v4)	One entry point: auth, budgets, allow-list, guardrails, logging, provider switching.
Stateless MCP (NEW v4)	Every request self-contained; scoped token checked per call; scales horizontally.
No long-lived keys (NEW v4)	Only the gateway holds provider keys; agents get short-lived scoped tokens.
Scale and release	Async with a semaphore, streaming, queues; canary, blue-green, shadow, flags.

Part B: Going beyond today

1. Self-hosting open models

Inference servers such as vLLM serve open models efficiently with continuous batching. Self-hosting makes sense with strict data rules, very high volume, or a need for custom models; budget for GPU capacity and operations.

2. Server-sent events

For browser chat UIs, servers often stream with server-sent events (text/event-stream), which browsers read easily and can reconnect automatically.

3. Back-pressure

When more requests arrive than you can serve, decide on purpose: queue them, return 429 with Retry-After, or degrade to a cheaper model. Never let them pile up silently until everything times out.

4. Regional deployment

For users in Japan and India, choose regions close to them and check that the models you need are available there. Data residency rules may decide the region for you.

5. Token exchange for agents

OAuth token exchange lets a gateway swap a user's token for a narrower, short-lived token that an agent or MCP server can use. The agent never sees the user's main credential or the provider key.

Part C: Check yourself

- When would you choose a managed agent runtime over your own loop?
- What four places should you check for data retention?
- What does an AI gateway centralise?
- Why does a stateless MCP server scale more easily?
- Why should an agent never hold a long-lived provider key?

Part D: Practice ideas for the weekend

- Add a /ready endpoint that returns 503 when a required secret is missing in prod.
- Write rollback criteria for SmartDesk's next release.
- Draw SmartDesk's architecture with the gateway, Key Vault and the MCP server.

Coming up on Day 35: observability, drift and optimisation, then the capstone from Tuesday.

Yoi shuumatsu wo (良い週末を): have a great weekend